# VolumetricSynth Midterm Presentation

progress update

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Spring 2026

Georgia Tech · College of Design
Center for
Music Technology

# Introduction

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- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.

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# Goals and Scope



- **Goal 1:** Core DSP blocks for volumetric synthesis.
- **Goal 2:** Playable plugin from MIDI to audio out.
- **Goal 3:** Simple UI for 3D mixing.
- **In scope:** Oscillator, filter, envelope, mixer, wiring, tests, UI state.
- **Out of scope:** Advanced FX, many presets, visual polish.

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# Architecture Overview

## Section Roadmap

- Codebase structure overview.
- Views: `Plugin`, `Visualization`, `Audio Engine`.
- Stack: `JUCE` + `OpenGL`.
- Per-view diagrams
- Finally, cross-module connections.

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## Plugin



- Host-facing plugin core and lifecycle.
- Main editor and UI composition root.
- Centralized parameter model (APVTS).
- **Flow:** host → processor → parameter/state bridge → audio + UI sync.

### Processor Class Card

| VolumetricSynthAudioProcessor |
| --- |
| +prepareToPlay(sampleRate, samplesPerBlock) |
| +processBlock(audioBuffer, midiBuffer) |
| +createEditor() |
| +parameterChanged(parameterID, newValue) |

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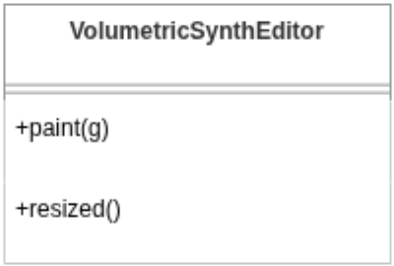
**Editor Class Card**



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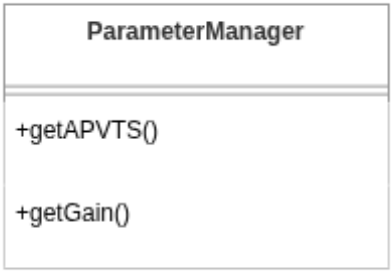
**Parameter Manager Class Card**



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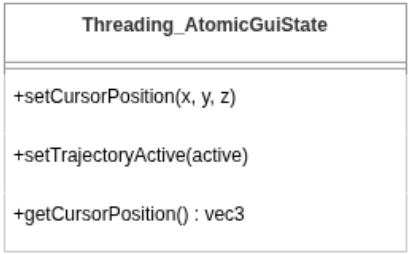
### Atomic GUI State Class Card

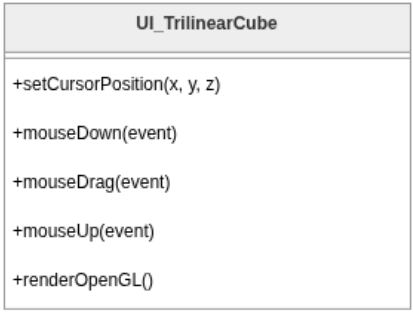


# Architecture Overview

## Visualization

- OpenGL cube view renders the 3D interaction space.
- Mouse drag drives cube rotation and cursor updates.
- Cursor position is normalized and pushed through a lock-free state bridge.
- **Flow:** user gesture → 3D view update → cursor state → DSP.

### 3D View Class Card



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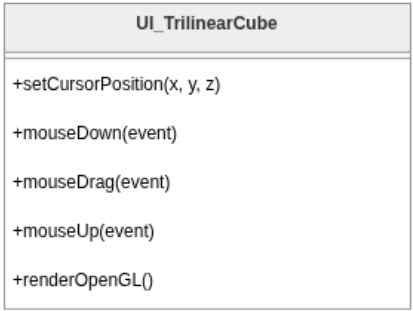
### Interaction Loop Class Card



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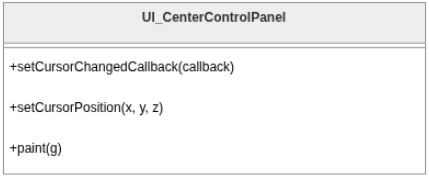
### Cursor State Class Card



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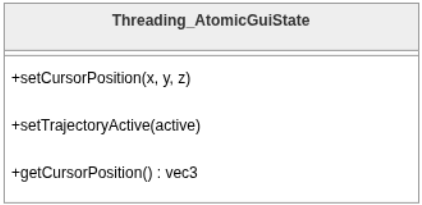
### Thread Bridge Class Card



# Architecture Overview

## Audio Engine

- MIDI and audio block orchestration.
- Voice allocation and note lifecycle.
- Per-voice synthesis signal path.
- **Flow:** MIDI/input → voice path → oscillator/mixer/envelope → output.

### Synth Engine Class Card

| Audio_SynthEngine |
| --- |
| +prepare(sampleRate, blockSize) |
| +processBlock(audioBuffer, midiBuffer) |
| +noteOn(channel, note, velocity) |
| +noteOff(channel, note, velocity) |

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### Voice Manager Class Card

| Audio_VoiceManager |
| --- |
| +prepare(sampleRate, blockSize) |
| +noteOn(channel, note, velocity) |
| +noteOff(channel, note, velocity) |
| +renderNextBlock(outputBuffer, startSample, numSamples) |

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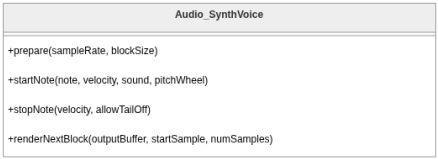

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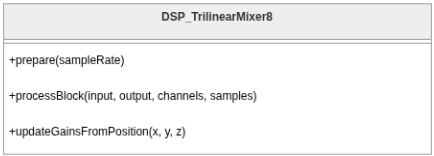
### DSP Stack Class Card



# Architecture Overview

## End-to-End Integration



- User interaction enters through the editor layer.
- Control and state propagate into real-time paths.
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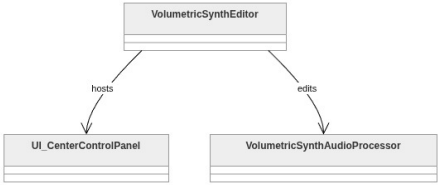
- **Flow:** UI → control/state bridge → processor → audio output.

**System Integration Overview**



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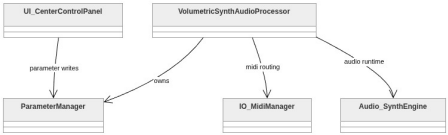
### Control Path Focus



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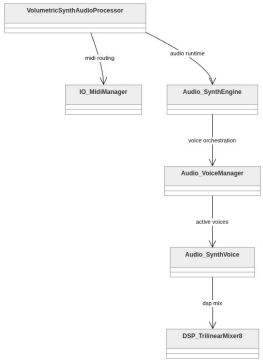
### Processor Coordination Focus



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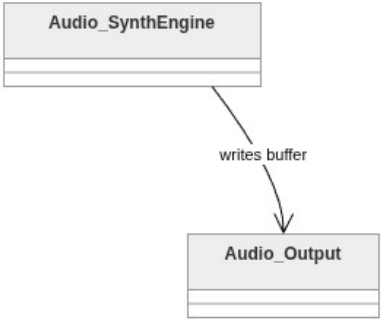

# Individual Responsibilities



- **Architecture/Integration (Task A):** plugin skeleton, module boundaries, build integration, and validation/QA ownership.
- **Audio Engine (Task B):** voice management, routing, and end-to-end realtime-safe orchestration.
- **DSP / Audio Effects (Task C):** DSP building blocks (e.g., mixer, filters), algorithm implementation and tuning.
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# Progress So Far

## Since Jan 20

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- Foundational DSP modules implemented (`DualFilter`, `TrilinearMixer8`, math helpers).
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- Tighten parameter mapping and edge-case behavior for UI and DSP interactions.
- Expand testing for integration-level scenarios (not only unit modules).
- Finalize end-to-end polish pass before submission.

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# Comprehensive Timeline (Jan 20 to May 5)



| Time | Focus | Owners | Done |
|---|---|---|---|
| Jan 20–Feb 2 | Kickoff + architecture | A + D | Build baseline OK |
| Feb 3–Feb 23 | DSP core modules | B | Module tests pass |
| Feb 24–Mar 16 | Plugin integration | A + C | Audio path works |
| Mar 17–Mar 31 | UI + state + tests | C + D | UI updates reliably |
| Apr 1–Apr 14 | Feature completion (1) | B + C | Core features done |
| Apr 15–Apr 24 | Optimization and Cleanup (2) | All | Stable integrated build |
| Apr 25–May 1 | Demo + docs + rehearsal | All | Script + slides ready |
| May 2–May 5 | Buffer + polish + submit | All | Submission package delivered |

# Demo



- video demo
- thanks, questions?

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